MetroSense Smart City Network

IoT NIDS Deployment — Build Manual

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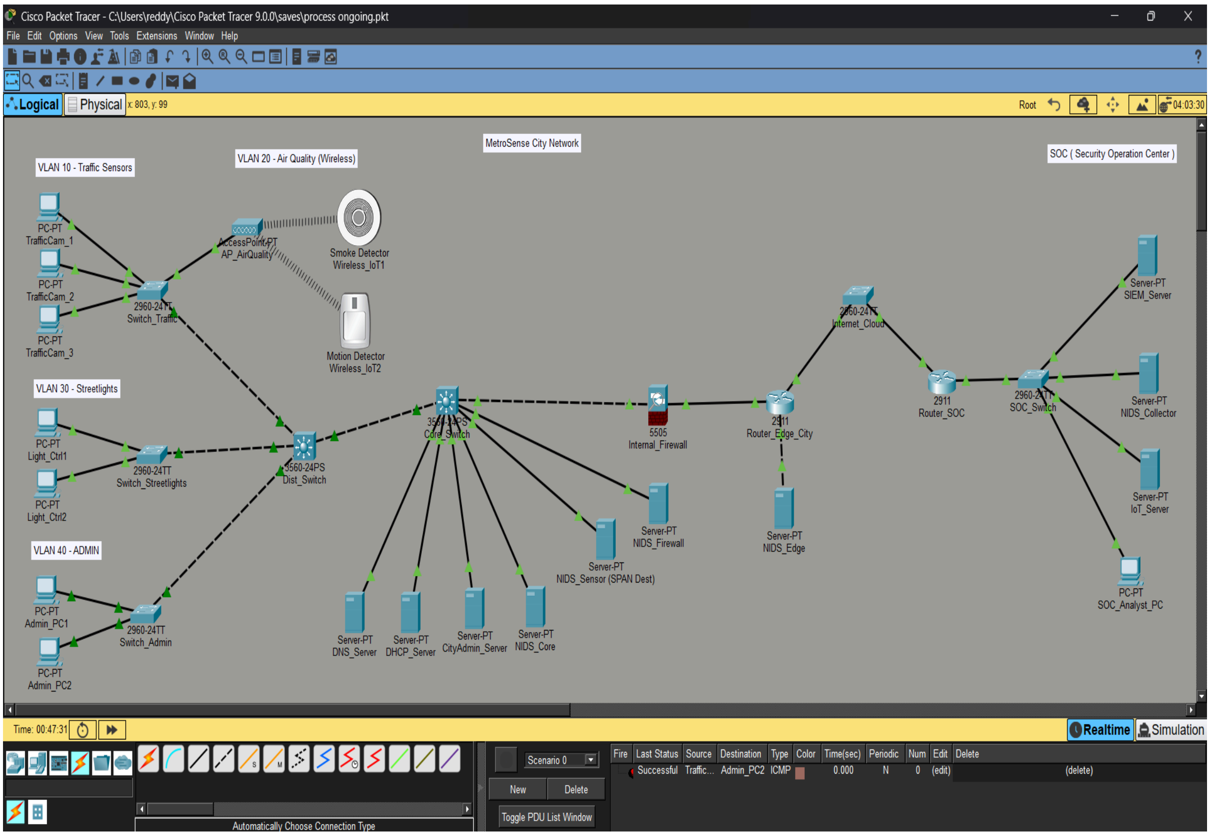
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1. Introduction

This manual documents the design and build process for MetroSense — a smart city network incorporating traffic sensors, air quality monitors, streetlight controllers, and city administration systems, protected by a layered Network Intrusion Detection System (NIDS) that reports to a remote Security Operations Centre (SentryOps SOC).

The project was built as a personal learning exercise in Cisco Packet Tracer, applying VLAN segmentation, SPAN-based traffic monitoring, firewalling, and site-to-site connectivity concepts to an IoT/smart-city scenario.

1.1 Objectives

- Segment city IoT device classes onto separate VLANs to contain the blast radius of a compromised device.
- Deploy a layered NIDS presence at the core, firewall, and edge of the network.
- Establish secure connectivity from the city network to a remote SOC.
- Produce a fully addressed, routable topology with verified end-to-end connectivity.

2. Network Topology Overview

The diagram below shows the complete MetroSense topology: four city-side VLANs, a wireless IoT cluster, a distribution/core switching layer, three NIDS monitoring points (Core, Firewall, Edge), and the link out to the SentryOps SOC.

3. Device Inventory and Naming

The table below lists every device in the project, its Packet Tracer default name, its assigned display name, and its role.

PT Default Name	Assigned Name	Role
PC0	TrafficCam_1	VLAN 10 endpoint — traffic camera sensor
PC1	TrafficCam_2	VLAN 10 endpoint — traffic camera sensor
PC2	TrafficCam_3	VLAN 10 endpoint — traffic camera sensor
AccessPoint0	AP_AirQuality	Wireless access point for VLAN 20 IoT sensors
(IoT) Wireless_IoT1	Smoke Detector	VLAN 20 wireless sensor
(IoT) Wireless_IoT2	Motion Detector	VLAN 20 wireless sensor (replaced faulty Temperature Monitor)
PC3	Light_Ctrl1	VLAN 30 endpoint — streetlight controller
PC4	Light_Ctrl2	VLAN 30 endpoint — streetlight controller
PC5	Admin_PC1	VLAN 40 endpoint — city ops admin PC
PC6	Admin_PC2	VLAN 40 endpoint — city ops admin PC
Switch0	Switch_Traffic	VLAN 10 access switch
Switch1	Switch_Streetlights	VLAN 30 access switch
Switch2	Switch_Admin	VLAN 40 access switch
Multilayer Switch0	Dist_Switch	Distribution layer switch
Multilayer Switch1	Core_Switch	Core layer switch (L3, SVI gateways, SPAN source)
Server0	DNS_Server	City DNS service
Server1	DHCP_Server	City DHCP service
Server2	CityAdmin_Server	City operations application server

Server3	NIDS_Core	NIDS sensor — core segment
Server4	NIDS_Sensor (SPAN Dest)	SPAN destination collecting mirrored core traffic
ASA0	Internal_Firewall	Internal firewall between core and edge (2 active interfaces: inside, outside)
Server5	NIDS_Firewall	NIDS sensor — relocated to Core_Switch/VLAN 50 (see Section 9)
Router0	Router_Edge_City	City edge router
Server6	NIDS_Edge	NIDS sensor — WAN-facing edge segment
Switch4 (2960-24TT)	Internet_Cloud	Simulated internet / WAN transit — a switch, not Cloud-PT (see Section 9)
Router1	Router_SOC	SOC edge router
Switch3	SOC_Switch	SOC internal switch
Server7	SIEM_Server	SOC — Security Information and Event Management server
Server8	NIDS_Collector	SOC — aggregates NIDS alerts from the city network
PC7	SOC_Analyst_PC	SOC analyst workstation

4. IP Addressing Plan

The network uses private RFC 1918 addressing for all city and SOC segments, with a point-to-point transit range representing the public-facing WAN link. Each VLAN and routed segment is a /24 except transit links, which are /30.

4.1 VLAN 10 — Traffic Sensors (192.168.10.0/24)

Device	IP Address	Subnet Mask	Gateway
Core_Switch VLAN 10 SVI (gateway)	192.168.10.1	255.255.255.0	—
TrafficCam_1	192.168.10.10	255.255.255.0	192.168.10.1
TrafficCam_2	192.168.10.11	255.255.255.0	192.168.10.1
TrafficCam_3	192.168.10.12	255.255.255.0	192.168.10.1

4.2 VLAN 20 — Air Quality / Wireless (192.168.20.0/24)

Device	IP Address	Subnet Mask	Gateway
Core_Switch VLAN 20 SVI (gateway)	192.168.20.1	255.255.255.0	—
AP_AirQuality (management)	192.168.20.10	255.255.255.0	192.168.20.1
Smoke Detector (Wireless_IoT1)	192.168.20.11	255.255.255.0	192.168.20.1
Motion Detector (Wireless_IoT2)	192.168.20.12	255.255.255.0	192.168.20.1

4.3 VLAN 30 — Streetlights (192.168.30.0/24)

Device	IP Address	Subnet Mask	Gateway

Core_Switch VLAN 30 SVI (gateway)	192.168.30.1	255.255.255.0	—
Light_Ctrl1	192.168.30.10	255.255.255.0	192.168.30.1
Light_Ctrl2	192.168.30.11	255.255.255.0	192.168.30.1

4.4 VLAN 40 — Admin (192.168.40.0/24)

Device	IP Address	Subnet Mask	Gateway
Core_Switch VLAN 40 SVI (gateway)	192.168.40.1	255.255.255.0	—
Admin_PC1	192.168.40.10	255.255.255.0	192.168.40.1
Admin_PC2	192.168.40.11	255.255.255.0	192.168.40.1

4.5 VLAN 50 — Core Server Segment (192.168.50.0/24)

Device	IP Address	Subnet Mask	Gateway
Core_Switch VLAN 50 SVI (gateway)	192.168.50.1	255.255.255.0	—
DNS_Server	192.168.50.10	255.255.255.0	192.168.50.1
DHCP_Server	192.168.50.11	255.255.255.0	192.168.50.1
CityAdmin_Server	192.168.50.12	255.255.255.0	192.168.50.1
NIDS_Core	192.168.50.13	255.255.255.0	192.168.50.1
NIDS_Sensor (SPAN Dest)	192.168.50.14	255.255.255.0	192.168.50.1
NIDS_Firewall (relocated — see Section 9)	192.168.50.15	255.255.255.0	192.168.50.1

4.6 Core-to-Firewall Transit Link

Device	IP Address	Subnet Mask	Gateway
Core_Switch (routed uplink)	192.168.55.1	255.255.255.252	—
Internal_Firewall (inside)	192.168.55.2	255.255.255.252	192.168.55.1

4.7 Firewall-to-Router Transit Link

Device	IP Address	Subnet Mask	Gateway
Internal_Firewall (outside)	192.168.65.1	255.255.255.252	—
Router_Edge_City (Gig0/0)	192.168.65.2	255.255.255.252	192.168.65.1

4.8 Edge / WAN-Facing Segment (192.168.70.0/24)

Device	IP Address	Subnet Mask	Gateway
Router_Edge_City (segment interface, gateway)	192.168.70.1	255.255.255.0	—
NIDS_Edge	192.168.70.10	255.255.255.0	192.168.70.1

4.9 WAN Link — City to SOC (203.0.113.0/30)

Device	IP Address	Subnet Mask	Gateway

Router_Edge_City (Gig0/1, toward Internet_Cloud)	203.0.113.1	255.255.255.252	—
Router_SOC (Gig0/0, toward Internet_Cloud)	203.0.113.2	255.255.255.252	—

4.10 SOC Internal Segment (192.168.100.0/24)

Device	IP Address	Subnet Mask	Gateway
Router_SOC (Fa0/1, gateway)	192.168.100.1	255.255.255.0	—
SIEM_Server	192.168.100.10	255.255.255.0	192.168.100.1
NIDS_Collector	192.168.100.11	255.255.255.0	192.168.100.1
SOC_Analyst_PC	192.168.100.12	255.255.255.0	192.168.100.1
Iot_Server	192.168.100.13	255.255.255.0	192.168.100.1

5. Step-by-Step Build Process

This section documents the full build order followed for this project, from empty canvas to a fully addressed, verified topology.

5.1 Step 1 — Place and Rename Devices

All end devices, switches, routers, the firewall, servers, the AP, and the cloud object were placed on the canvas first, then renamed using the naming scheme in Section 3 (double-click each device's label to rename).

5.2 Step 2 — Cable VLAN 10 (Traffic Sensors)

1. Select the cable tool and choose "Automatically Choose Connection Type".
2. Connect TrafficCam_1, TrafficCam_2, and TrafficCam_3 to Switch_Traffic (one link per device).

5.3 Step 3 — Cable VLAN 20 (Air Quality, Wireless)

3. Connect AP_AirQuality to Switch_Traffic using a wired copper link.
4. Attempted to use Laptop-PT devices as wireless sensors; this failed because default laptops in this PT build do not carry a compatible wireless NIC (WPC300N is a desktop-only module).
5. Replaced the laptops with native wireless IoT devices from the IoT/Components category: Smoke Detector and Motion Detector. These associate wirelessly to the AP without any module installation.
6. On AP_AirQuality: set SSID to MetroSense_IoT, Authentication to WPA2-PSK, and a shared passphrase.
7. On each IoT sensor's config, set the matching SSID and passphrase so they associate automatically.

5.4 Step 4 — Cable VLAN 30 (Streetlights) and VLAN 40 (Admin)

8. Connect Light_Ctrl1 and Light_Ctrl2 to Switch_Streetlights.
9. Connect Admin_PC1 and Admin_PC2 to Switch_Admin.

5.5 Step 5 — Uplink Access Switches to Distribution

10. Connect Switch_Traffic, Switch_Streetlights, and Switch_Admin to Dist_Switch.
11. Auto-connect selects copper straight-through for these links; Auto-MDIX on the 2960/3560 platforms makes this equivalent to a crossover cable, so either works.

5.6 Step 6 — Distribution to Core, and Core Servers

12. Connect Dist_Switch to Core_Switch.
13. Connect DNS_Server, DHCP_Server, CityAdmin_Server, NIDS_Core, and NIDS_Sensor to Core_Switch (five servers, as planned).

5.7 Step 7 — Firewall

14. Connect Core_Switch to Internal_Firewall (ASA 5505).
15. Initially connected NIDS_Firewall to a third Ethernet interface on Internal_Firewall. The ASA 5505 Base license rejected this with 'ERROR: This license does not allow configuring more than 2 interfaces with nameif'. Resolved by relocating NIDS_Firewall to Core_Switch (VLAN 50) instead, keeping the ASA to exactly two active interfaces (inside, outside). See Section 9 for full details.

5.8 Step 8 — Edge Router and Edge NIDS

1. Connect Internal_Firewall to Router_Edge_City.
2. Connect NIDS_Edge to a free interface on Router_Edge_City.

5.9 Step 9 — WAN Link to SOC

3. Initially attempted a Cloud-PT object (Internet_Cloud) between Router_Edge_City and Router_SOC, including fitting a PT-CLOUD-NM-1CFE module for a second Ethernet port and setting both ports to the same internal 'Cable' provider network. Traffic still failed to pass end to end (0% ping success) despite correct IP addressing and routing on both routers.
4. Resolved by deleting the Cloud-PT object entirely and replacing it with a plain 2960-24TT switch (kept the display name Internet_Cloud for diagram consistency). No configuration was required on the switch; connectivity worked immediately. See Section 9.

5.10 Step 10 — SOC-Side Connections

5. Connect Router_SOC to SOC_Switch.
6. Connect SIEM_Server, NIDS_Collector, and SOC_Analyst_PC to SOC_Switch.

6. Device Configuration

6.1 VLAN Creation (Dist_Switch and Core_Switch)

```
enable
configure terminal
vlan 10
 name Traffic_Sensors
exit
vlan 20
```

```
 name Air_Quality
exit
vlan 30
 name Streetlights
exit
vlan 40
 name Admin
exit
vlan 50
 name Core_Servers
exit
```

6.2 Access Port Assignment

On Switch_Traffic:

```
enable
configure terminal
interface range fa0/1 - 3
 switchport mode access
 switchport access vlan 10
exit
interface fa0/4
 switchport mode access
 switchport access vlan 20
exit
```

On Switch_Streetlights:

```
enable
configure terminal
interface range fa0/1 - 2
 switchport mode access
 switchport access vlan 30
exit
```

On Switch_Admin:

```
enable
configure terminal
interface range fa0/1 - 2
 switchport mode access
 switchport access vlan 40
exit
```

6.3 Trunk Links

On each access switch, the uplink port toward Dist_Switch:

```
interface fa0/x
 switchport mode trunk
```

On Dist_Switch, matching downlink ports plus the uplink to Core_Switch:

```
interface range fa0/1 - 4
 switchport mode trunk
exit
interface fa0/7
 switchport mode trunk
```

Replace port numbers with the actual interfaces used in the build — refer to the port labels visible in Logical view.

6.4 Core_Switch — SVI Gateways and Routing

```
enable
configure terminal
ip routing
interface vlan 10
 ip address 192.168.10.1 255.255.255.0
 no shutdown
exit
interface vlan 20
 ip address 192.168.20.1 255.255.255.0
 no shutdown
exit
interface vlan 30
 ip address 192.168.30.1 255.255.255.0
 no shutdown
exit
interface vlan 40
 ip address 192.168.40.1 255.255.255.0
 no shutdown
exit
interface vlan 50
 ip address 192.168.50.1 255.255.255.0
 no shutdown
exit
```

6.5 Core_Switch — Routed Uplink to Firewall

```
interface fa0/x
 no switchport
 ip address 192.168.55.1 255.255.255.252
 no shutdown
exit
ip route 192.168.60.0 255.255.255.0 192.168.55.2
ip route 192.168.70.0 255.255.255.0 192.168.55.2
ip route 192.168.100.0 255.255.255.0 192.168.55.2
```

6.6 Internal_Firewall (ASA 5505) — Final Working Configuration

The ASA 5505 Base license permits a maximum of two active (nameif'd) interfaces. The original three-zone design (inside / firewall-segment / outside) was rejected by the device with a licensing error, so the final design uses exactly two: inside and outside. NIDS_Firewall was moved to the core server segment instead (Section 4.5).

A fresh ASA also ships with factory-default nameif assignments already applied to Vlan1 ("inside") and Vlan2 ("outside"). Because a nameif value can only be assigned to one interface at a time, these defaults must be cleared before assigning the same names to the VLANs actually carrying traffic (Vlan10 and Vlan30 in this build) — otherwise the command is silently rejected or, worse, the name ends up active on the wrong, unused interface. Both mistakes were made and corrected during this build; the sequence below reflects the corrected, final process.

```
enable
configure terminal

! Physical ports mapped to VLANs
interface Ethernet0/0
 switchport access vlan 10
exit
interface Ethernet0/1
 switchport access vlan 30
exit

! Clear factory-default names before assigning new ones
interface vlan1
 no nameif
exit
interface vlan2
 no nameif
exit

! Inside interface
interface Vlan10
 nameif inside
 security-level 100
 ip address 192.168.55.2 255.255.255.252
 no shutdown
exit

! Outside interface
interface Vlan30
 nameif outside
 security-level 0
 ip address 192.168.65.1 255.255.255.252
 no shutdown
exit

! Routes
route inside 192.168.10.0 255.255.255.0 192.168.55.1
route inside 192.168.20.0 255.255.255.0 192.168.55.1
route inside 192.168.30.0 255.255.255.0 192.168.55.1
route inside 192.168.40.0 255.255.255.0 192.168.55.1
route inside 192.168.50.0 255.255.255.0 192.168.55.1
route outside 0.0.0.0 0.0.0.0 192.168.65.2
```

```
! ICMP inspection so replies to through-traffic pings are permitted back
class-map inspection_default
 match default-inspection-traffic
policy-map global_policy
 class inspection_default
  inspect icmp
service-policy global_policy global

! icmp permit (a real-ASA command) is not supported in this Packet
! Tracer version; an access-list achieves the same practical effect
access-list ICMP_IN extended permit icmp any any
access-group ICMP_IN in interface inside
access-group ICMP_IN in interface outside
```

6.7 Router_Edge_City

```
enable
configure terminal
interface gigabitEthernet0/0
 ip address 192.168.65.2 255.255.255.252
 no shutdown
exit
interface fastEthernet0/0
 ip address 192.168.70.1 255.255.255.0
 no shutdown
exit
interface gigabitEthernet0/1
 ip address 203.0.113.1 255.255.255.252
 no shutdown
exit
ip route 192.168.10.0 255.255.255.0 192.168.65.1
ip route 192.168.20.0 255.255.255.0 192.168.65.1
ip route 192.168.30.0 255.255.255.0 192.168.65.1
ip route 192.168.40.0 255.255.255.0 192.168.65.1
ip route 192.168.50.0 255.255.255.0 192.168.65.1
ip route 192.168.100.0 255.255.255.0 203.0.113.2
```

6.8 Router_SOC

```
enable
configure terminal
interface gigabitEthernet0/0
 ip address 203.0.113.2 255.255.255.252
 no shutdown
exit
interface gigabitEthernet0/1
 ip address 192.168.100.1 255.255.255.0
 no shutdown
exit
ip route 192.168.10.0 255.255.255.0 203.0.113.1
ip route 192.168.20.0 255.255.255.0 203.0.113.1
ip route 192.168.30.0 255.255.255.0 203.0.113.1
ip route 192.168.40.0 255.255.255.0 203.0.113.1
```

```
ip route 192.168.50.0 255.255.255.0 203.0.113.1
ip route 192.168.70.0 255.255.255.0 203.0.113.1
```

6.9 End-Device IP Configuration

For every PC and server, open the device's Desktop tab → IP Configuration, and enter the Static IP address, Subnet Mask, and Default Gateway from the tables in Section 4. Wireless IoT sensors are configured via their own Config tab with the same addressing plus the matching SSID/passphrase set out in Step 3 (Section 5.3).

7. NIDS Monitoring Configuration (SPAN)

NIDS_Sensor receives a mirrored copy of all traffic traversing Core_Switch via a SPAN (Switched Port Analyzer) session, so it can passively inspect traffic without sitting inline.

```
enable
configure terminal
monitor session 1 source vlan 10 - 50
monitor session 1 destination interface fa0/8
```

Confirmed working configuration (show monitor session 1):

```
Session 1
---------
Type              : Local Session
Source VLANs      :
        Both      : 10,11,...,49
Destination Ports : Fa0/8
        Encapsulation : Native
        Ingress       : Disabled
```

8. Verification and Testing

Connectivity was verified in stages, from local segment outward to the SOC, with each stage confirmed before moving to the next:

1. Local VLAN gateway reachability, e.g. from Admin_PC1: ping 192.168.10.1 / 192.168.50.1 — confirmed working.
2. Through the firewall to the core server segment: ping 192.168.50.15 (NIDS_Firewall) — confirmed working after the ASA nameif/license fixes in Section 9.
3. To the edge segment beyond the firewall: ping 192.168.70.10 (NIDS_Edge) — confirmed working, 100% success on repeat attempts (an initial single dropped packet on first contact is expected ARP-resolution behaviour, not a fault).
4. Router-to-router across the WAN link: ping 203.0.113.2 from Router_Edge_City — failed while Internet_Cloud (Cloud-PT) was in place; succeeded at 100% immediately after replacing it with a plain switch.

5. Full city-to-SOC path from Admin_PC1: ping 192.168.100.10, .11, .12 (SIEM_Server, NIDS_Collector, SOC_Analyst_PC) — confirmed working end to end.
6. SPAN session verified active on Core_Switch via show monitor session 1 — Source VLANs 10-50, Destination Fa0/8.
7. Use Simulation mode to trace a single ping packet hop by hop and confirm it follows the expected path: access switch → distribution → core → firewall → edge router → Internet_Cloud (switch) → SOC router → SOC switch.

9. Issues Encountered and Resolutions

This build surfaced a number of Packet Tracer-specific behaviours and a genuine ASA licensing limit. Each is documented below with root cause and fix, since diagnosing them was a significant part of the practical work.

Issue	Root Cause / Resolution
Laptop wireless module (WPC300N) reported incompatible	WPC300N is a desktop-PC-only module, not valid for laptops. Resolved by replacing the laptops entirely with native wireless IoT devices (Smoke Detector, Motion Detector) from the IoT/Components category, which associate to the AP without any module installation.
Temperature Monitor would not associate to the AP	Config mismatch could not be isolated. Replaced the device with a Motion Detector, copying the exact SSID/passphrase from the already-working Smoke Detector.
3560 multilayer switches rejected switchport mode trunk	The 3560 platform requires the trunk encapsulation to be set explicitly before trunk mode is allowed (unlike the 2960). Fixed with switchport trunk encapsulation dot1q before switchport mode trunk, on both Dist_Switch and Core_Switch.
ASA rejected 'nameif inside' on Ethernet0/x directly	ASA 5505 physical ports are Layer 2 by default; nameif/ip address/security-level can only be applied to a VLAN interface. Fixed by assigning each physical port to a VLAN (switchport access vlan X) and configuring nameif on the corresponding interface vlan X instead.
ASA rejected a third nameif'd interface (firewall-segment)	The ASA 5505 Base license permits a maximum of 2 active interfaces. Resolved by removing the third zone entirely: NIDS_Firewall was relocated to Core_Switch on the existing VLAN 50 server segment, leaving the ASA with exactly inside and outside.
ICMP through the firewall worked one direction but not the return	By default the ASA does not track ICMP as a stateful session, so echo replies for through-traffic were silently dropped. Fixed by adding inspect

	icmp under policy-map global_policy and applying it globally with service-policy global_policy global.
Pings to the ASA's own inside/outside interface IPs still failed after ICMP inspection was added	inspect icmp only covers traffic passing through the firewall, not traffic destined to the firewall itself. The real-ASA command icmp permit any inside/outside is not implemented in this Packet Tracer version (rejected as invalid input regardless of prompt level). Worked around with an access-list applied inbound on both interfaces: access-list ICMP_IN extended permit icmp any any, then access-group ICMP_IN in interface inside/outside.
A freshly added ASA silently kept nameif inside/outside on the factory-default Vlan1/Vlan2 instead of the intended Vlan10/Vlan30	A brand-new ASA 5505 ships with nameif inside pre-assigned to Vlan1 and dhcp-based nameif outside pre-assigned to Vlan2. These must be explicitly cleared (interface vlan1 / no nameif, same for vlan2) before assigning the same names to the VLANs that actually carry traffic — otherwise the active zone silently sits on the wrong, unused interface with no error at all.
Internet_Cloud (Cloud-PT) passed 0% of traffic between Router_Edge_City and Router_SOC despite correct IPs, routes, and firewall rules on both sides	Cloud-PT requires each connected port to be explicitly associated with an internal 'provider network' (Cable/DSL/etc.) for traffic to bridge between them; even after setting both ports to the same provider network, traffic still did not pass reliably. Resolved by deleting Internet_Cloud entirely and replacing it with a plain 2960-24TT switch (kept the same display name for diagram consistency) — no configuration was needed and connectivity worked immediately.
Router-to-router ping across the WAN link failed even though each router's own LAN and routing table were confirmed correct	This isolated the fault to the transit device (Internet_Cloud) rather than either router's configuration — a useful diagnostic pattern: when both endpoints check out individually but the path between them fails, suspect the device in between.
Cable-type confusion for switch-to-switch and switch-to-router links	Resolved by using the 'Automatically Choose Connection Type' cable tool throughout, which relies on Auto-MDIX on the 2960/3560/ASA platforms and removes the need to manually pick straight-through vs crossover.
Red status triangles on newly cabled links before IP configuration	Expected behaviour — links show down until IP addressing and no shutdown are applied to both ends; not a cabling fault.

10. Summary

This manual has taken the MetroSense topology from an empty canvas through device placement, cabling, VLAN segmentation, IP addressing, routing, firewall configuration, SPAN-based NIDS deployment, and end-to-end verification. The result is a segmented smart-city network with layered intrusion detection reporting to a remote SOC over a routed WAN link — a working demonstration suitable for a portfolio write-up or resume project.